

Flowchart / System Design Notes

- Flowcharts are used to visually represent the workflow and architecture of a system before coding begins.
- Professional software development usually starts with planning and system design before implementation.
- Standard flowchart symbols:
 - Oval = Start/End
 - Rectangle = Process/Task
 - Diamond = Decision/Question
 - Arrows = Direction of workflow
- Validation and error-handling steps are important in professional workflows.
- Each module should be tested separately before integrating the full system.
- Important sections for my project:
 - API connection
 - Data collection
 - Data cleaning/preprocessing
 - Ticker extraction
 - Sentiment analysis
 - Analytics engine
 - Dashboard visualization
- Flowcharts should clearly show data movement between components.
- Gantt charts help organize tasks, deadlines, and project phases.

Twitter/X API Research Notes

- Twitter/X API can collect tweets, timestamps, hashtags, likes, reposts, and engagement metrics.
- API authentication requires developer account access and API keys.
- Tweepy is a Python library commonly used for Twitter API integration.
- Tweets are unstructured text data and require preprocessing before analysis.
- Possible stock-related keywords/tickers:
 - TSLA
 - NVDA
 - AAPL
 - AMZN
- API rate limits may restrict the amount of data collected at once.
- JSON responses from APIs need to be converted into structured formats such as pandas DataFrames.

Sentiment Analysis Notes

- Sentiment analysis classifies text as positive, negative, or neutral.
- NLP preprocessing steps may include:
 - lowercasing text
 - removing symbols/emojis
 - removing stopwords
 - tokenization
- Sentiment scores can help identify overall market opinion on specific stocks.
- High mention frequency combined with positive sentiment may indicate trending stocks.
- Possible outputs for dashboard:
 - sentiment trend graph
 - ticker rankings
 - mention volume
 - positive vs negative sentiment counts

GitHub / Project Organization Notes

- Repository should be public and organized professionally.
- README file should explain:
 - project purpose
 - folder structure
 - how to run the code
 - dependencies/libraries
- Important folders:
 - data/
 - scripts/
 - dashboard/
 - docs/
- Code should contain comments for readability and reproducibility.
- GitHub commits should be consistent throughout the internship.